

I'll keep **operate** at the same fullest-reference standard: broad family sweep first, then specialist branches, fixed constructions, distinctions, and only at the end consolidate the database cores.

Word family: **operate / operates / operated / operating / operation / operations / operator / operational / operationally / operative / operatively / operable / inoperable / operability**

Base word: operate

Past tense: operated

Past participle: operated

-ing form: operating

Major related forms:

- * operation
- * operations
- * operator
- * operational
- * operationally
- * operative
- * operatively
- * operable
- * inoperable
- * operability

Important extended forms:

- * operationalise / operationalize
- * operationalisation / operationalization
- * preoperative / pre-operative
- * postoperative / post-operative
- * perioperative
- * non-operative / nonoperative
- * interoperable
- * interoperability

Major compounds/constructions:

- * operate a machine
- * operate a business
- * operate a service
- * operate in/from/under/as
- * operate at a profit/loss
- * operate at capacity
- * operate on a patient
- * operating costs
- * operating expenses

- * operating profit/income/margin
- * operating system
- * operating room/theatre
- * business operations
- * military/police operation
- * mathematical operation
- * operator
- * operational readiness
- * operational definition
- * operative word
- * operable/inoperable tumour

****Part of speech:** verb (動詞)**

Your sentence:

> “...the private tutoring industry ****externalise its operating costs**** while avoiding responsibility.”

Meaning in this passage

Here, ****operating costs**** means ****the ongoing costs required to run a business or service in its normal day-to-day activities****.

For a tutoring business, these may include things such as:

- * premises
- * utilities
- * staff
- * equipment
- * administration
- * cleaning
- * other recurring running expenses

The writer argues that tutors using restaurants for lessons avoid some of these costs and effectively shift part of them onto the restaurant.

****繁體中文： ** **營運成本；日常經營所需的成本****

> “externalise its ****operating costs****”

= 「把本應由自己承擔的****營運成本轉嫁給外界 / 他人****」

So the contextual family sense is:

> ****operate a business = 經營 / 營運業務****

> ****operating costs = 經營該業務所產生的日常成本****

A. OPERATE = FUNCTION / WORK

1. **operate = function; work (運作) — 運作 ; 運行**

One of the most fundamental meanings.

Practice 1

> “The machine is **operating** normally.”

「機器正正常**運作**。」

Practice 2

> “The system can **operate** without an internet connection.”

「這套系統可以在沒有網絡連線的情況下**運作**。」

Here **operate** is intransitive:

> something itself functions.

2. **operate properly / normally / efficiently**

Common:

- * operate properly
- * operate normally
- * operate efficiently
- * operate safely
- * operate reliably

Practice 3

> “The equipment cannot operate safely at this temperature.”

「設備在這個溫度下無法安全**運作**。」

3. **operate at + temperature / pressure / speed / voltage**

Technical:

> “The device operates at 12 volts.”

「裝置以 12 伏特**運作**。」

> “The engine operates at high temperature.”

「引擎在高溫下**運轉**。」

4. **operate within a range**

> “The sensor operates within a limited temperature range.”

「感應器只能在有限的溫度範圍內**運作**。」

5. **operating condition(s) = 運作條件**

Examples:

- * normal operating conditions
- * extreme operating conditions
- * safe operating conditions

6. **operating range = 運作範圍**

May describe:

- * temperature
- * voltage
- * pressure
- * speed
- * frequency

within which equipment is intended to function.

B. OPERATE = CONTROL A MACHINE

7. **operate something = control/use a machine or device (操作) — 操作；操控**

This is the major transitive machinery sense.

Practice 4

> “Only trained staff may **operate** this machine.”

「只有受過訓練的員工才可**操作**這部機器。」

Practice 5

> “Do you know how to operate the equipment?”

「你知道怎樣**操作**這套設備嗎？」

8. **operate machinery = 操作機械**

Common:

* operate heavy machinery

* operate construction equipment

* operate industrial equipment

9. **operate controls = 操作控制裝置**

> “The driver operates the controls from the cab.”

「駕駛員在駕駛室內**操作控制裝置**。」

10. **manually operated = 手動操作的**

> “The door is manually operated.”

「這扇門是**手動操作**的。」

11. **electrically / hydraulically / pneumatically operated**

Technical:

- * electrically operated = 電動操作
- * hydraulically operated = 液壓操作
- * pneumatically operated = 氣動操作

Here **operated** means:

> actuated/controlled by that mechanism.

C. MACHINE OPERATES VS PERSON OPERATES MACHINE

Critical grammar distinction:

- > **The machine operates.**
- > 機器運作。

VS

- > **She operates the machine.**
- > 她操作機器。

Same verb, different argument structure.

D. OPERATE = RUN / MANAGE A BUSINESS

12. **operate a business = run/manage a business (經營) — 經營 ; 營運**

This is the branch behind your passage.

Practice 6

> “They **operate** a small tutoring business.”

「他們**經營**一間小型補習公司。」

Practice 7

> “The family has operated the restaurant for twenty years.”

「這個家庭已**經營**這間餐廳二十年。」

13. **operate a company = 經營公司**

Formal/business style.

> “The company operates several subsidiaries.”

Can mean:

> 管理 / 營運數家子公司。

14. **operate a shop / hotel / restaurant / school**

> operate a hotel

> = 經營酒店

> operate a restaurant

> = 經營餐廳

15. **owner-operated = 由業主親自經營**

> an owner-operated restaurant

「由業主親自**經營**的餐廳」

E. OPERATE = PROVIDE / RUN A SERVICE

16. **operate a service = run/provide it (營辦) **

Practice 8

> “The company operates a free shuttle service.”

「公司**營辦**免費接駁服務。」

17. **operate a bus/train/air service**

Can mean:

> organize/provide the service

not necessarily personally drive the vehicle.

> “The airline operates three flights a day.”

「航空公司每日**營運 / 提供**三班航班。」

18. **a route operates / service operates**

Intransitive:

> “The service operates from Monday to Friday.”

「服務逢星期一至五**運作 / 提供服務**。」

19. **operate between A and B**

Transport:

> “Buses operate between the station and the airport.”

「巴士在車站和機場之間**行駛 / 提供服務**。」

20. **operate 24 hours a day**

> “The service operates 24 hours a day.”

「這項服務每日**24 小時運作**。」

F. OPERATE = CONDUCT ACTIVITIES IN A PLACE / MARKET

21. **operate in + place/industry = carry on activities there (營運 ; 活動) **

Practice 9

> “The company operates in several countries.”

「公司在數個國家**營運**。」

22. **operate in a market / sector**

> “The firm operates in the healthcare sector.”

「該公司在醫療行業**經營業務**。」

23. **operate internationally / globally**

> “The organization operates globally.”

「該機構在全球各地**運作 / 開展活動**。」

24. **operate locally**

= 在本地營運 / 活動。

25. **operate from + location**

Practice 10

> “The company operates from an office in the city centre.”

「公司以市中心的一間辦公室作為**營運基地**。」

Meaning:

> use that place as the base of operations.

G. OPERATE UNDER / AS / WITHIN

26. **operate under a name = trade/work using that name (以……名義營運) **

> “The business operates under a different name.”

「這間企業以另一個名稱**經營**。」

27. **operate under a licence/license = 持牌營運**

> “The centre must operate under a valid licence.”

「中心必須持有效牌照**營運**。」

28. **operate under rules / regulations**

> “All operators must operate under the same rules.”

「所有營辦者都必須按照相同規則**營運**。」

29. **operate as + role/type**

Practice 11

> “The building operates as a community centre.”

「這棟建築物**作為社區中心運作**。」

30. **operate within the law = 依法營運**

> “Businesses must operate within the law.”

「企業必須**依法經營**。」

31. **operate outside the law = 在法律規範以外 / 非法活動**

Context determines whether it means technically unlawful or merely outside a particular regulatory structure.

H. OPERATE AT A PROFIT / LOSS

32. **operate at a profit = operate profitably (有盈利地營運) **

> “The branch now operates at a profit.”

「這家分店現在已能**盈利營運**。」

33. **operate at a loss = operate while losing money (虧損經營) **

Practice 12

> “The service has operated at a loss for years.”

「這項服務多年來一直**虧損營運**。」

34. **operate at full capacity = 滿負荷 / 滿容量運作**

> “The hospital is operating at full capacity.”

「醫院正以**滿負荷狀態運作**。」

35. **operate below capacity = 未達最大容量運作**

I. **OPERATING COSTS** — EXACT PASSAGE FAMILY

36. **operating cost = cost of running normal activities (營運成本) **

Practice 13

> “Rent is a major operating cost.”

「租金是一項主要的**營運成本**。」

Examples can include:

- * wages
- * utilities
- * maintenance
- * premises costs
- * consumables
- * routine administration

depending on the business/accounting definition being used.

37. ****operating costs = ongoing costs of running a business****

This is your passage:

> tutoring centres should normally bear costs associated with having teaching premises.

The criticism is that restaurant-based tutoring lets them avoid part of those costs.

38. ****fixed operating costs = 固定營運成本****

Costs that do not change much with short-term activity levels, within the relevant range.

Examples may include:

- * rent
- * some salaries
- * insurance

depending on accounting treatment.

39. ****variable operating costs = 變動營運成本****

Costs changing with level of output/activity.

40. ****operating cost vs overhead****

Important given your earlier target ****overhead****.

**operating costs**

broad category of costs incurred in normal business operation.

overhead

generally indirect costs that cannot conveniently be traced to a specific unit/product/service.

Therefore:

> overhead may form part of operating costs,

but:

> operating costs ≠ all overhead only.

41. **operating cost vs capital cost**

operating cost / operating expenditure

cost of running activities.

capital expenditure

cost of acquiring/improving longer-term assets.

Example:

> electricity bill → operating expense

> buying a new building → generally capital expenditure

Exact accounting treatment can depend on standards/context.

J. **OPERATING EXPENSE / OPEX**

42. **operating expense = expense incurred through ordinary operations (營運開支)**

Common abbreviation:

> **OPEX / OpEx**

Examples:

- * salaries
- * rent
- * utilities
- * maintenance

depending on accounting classification.

43. **operating expenses vs cost of goods sold**

Financial reporting often distinguishes:

- * cost of goods sold / cost of sales
- * operating expenses

but classification varies by industry/accounting framework.

So do not automatically treat every operating cost as OPEX in every statement presentation.

K. FINANCE / ACCOUNTING FAMILY

44. **operating income = 營業利益 / 營運收益**

Broadly:

> profit generated from ordinary business operations before certain non-operating items.

Exact presentation depends on accounting framework.

45. **operating profit = 營業利潤**

Often similar to **operating income**, though usage/accounting presentation can differ.

46. **operating margin = 營業利潤率**

Broadly:

> operating profit ÷ revenue

under the relevant definitions.

47. ****operating loss = 營業虧損****

Normal operations generated a loss.

48. ****operating revenue = 營運 / 營業收入****

Revenue from primary operating activities.

49. ****operating cash flow = 營運現金流****

Cash flows associated with operating activities.

Often shown as:

> cash flow from operating activities.

50. ****operating activities = 營運活動****

One major category in a cash-flow statement, alongside investing and financing activities under many accounting frameworks.

51. ****operating budget = 營運預算****

Budget for normal day-to-day operations.

52. ****operating leverage = 營運槓桿****

Financial/business concept concerning how fixed operating costs influence sensitivity of operating profit to changes in sales/activity.

53. **operating cycle = 營運周期**

Broadly, the cycle from acquiring inputs/inventory through sale and collection of cash/receivables, under the accounting/business definition.

54. **non-operating income/expense = 非營運收入 / 支出**

Items not arising from primary ordinary business operations under the applicable presentation.

L. OPERATIONS = DAY-TO-DAY BUSINESS ACTIVITY

55. **operations = activities involved in running an organization (營運) **

Practice 14

> “She oversees the company's daily **operations**.”

「她負責監督公司的日常**營運**。」

56. **business operations = 業務營運**

Can include:

- * production
- * logistics
- * staffing
- * service delivery
- * administration

57. **operations department = 營運部門**

Department responsible for ongoing operational activities.

58. **operations manager = 營運經理**

Person managing processes/day-to-day delivery.

59. **operations management = 營運管理**

Management discipline concerned with producing/delivering goods and services efficiently.

M. OPERATION = PROCESS / WORKING OF A SYSTEM

60. **operation = action/process of functioning (運作)**

Practice 15

> “The manual explains the operation of the machine.”

「說明書解釋機器的**運作方式**。」

61. **principle of operation = 運作原理**

> “The diagram shows the principle of operation.”

「圖表顯示其**運作原理**。」

62. **in operation = functioning/in use (運作中)**

Practice 16

> “The system is now in operation.”

「系統現在已經**投入運作**。」

63. **put something into operation = 使投入運作**

> “The new system will be put into operation next month.”

「新系統下月將**投入運作**。」

64. **come into operation = begin functioning / take effect**

Can describe:

machine/system

開始運作

law/rule

開始生效

65. **cease operation(s) = stop functioning/business activity (停止運作 / 營運) **

> “The factory ceased operations.”

「工廠**停止營運**。」

N. OPERATION = ORGANIZED ACTIVITY

66. **operation = organized activity undertaken for a purpose (行動 / 作業) **

Practice 17

> “Police launched a major search operation.”

「警方展開大型**搜查行動**。」

Broad enough for:

* search operation

* rescue operation

* relief operation

* recovery operation

67. **rescue operation = 救援行動**

68. **search-and-rescue operation = 搜救行動**

69. **relief operation = 救援 / 賑災行動**

70. **cleanup operation = 清理行動**

71. **smuggling operation / criminal operation**

> organized activity/business engaged in illegal conduct.

Here **operation** can mean:

> 組織 / 活動網絡.

72. **a small operation = small business/organization (小型營運單位) **

Informal business sense:

> “They run a small catering operation.”

「他們經營一個小型餐飲**業務 / 營運單位**。」

O. MEDICINE — SURGERY

73. **operate on someone = perform surgery (動手術) **

Practice 18

> “The surgeon operated on the patient.”

「外科醫生為病人**動手術**。」

The verb normally uses:

> **operate on + patient/body part**

not:

> *operate the patient* ❌

74. **operate on a body part**

> “Doctors operated on his knee.”

「醫生為他的膝部**進行手術**。」

75. **operation = surgical procedure (手術)**

Practice 19

> “She had an operation on her knee.”

「她的膝部接受了**手術**。」

Common:

- * have an operation
- * undergo an operation
- * perform an operation

76. **major/minor operation**

= 大型 / 小型手術

The distinction depends on medical context and is not always formally standardized by the words alone.

77. **emergency operation = 緊急手術**

78. **elective operation = 非緊急預約手術 / 選擇性手術**

Medical meaning:

> planned rather than immediately emergency-required.

It does ****not**** necessarily mean “optional” in the ordinary casual sense.

P. OPERATING ROOM / THEATRE

79. ****operating room = 手術室****

Especially AmE/international medical English.

80. ****operating theatre/theater = 手術室****

Common especially in BrE and related varieties.

81. ****operating table = 手術台****

82. ****operating surgeon = 主刀 / 進行手術的外科醫生****

Q. OPERATIVE — MEDICAL

83. ****operative = surgical (手術的) ****

Examples:

* operative treatment

* operative procedure

* operative management

****繁體中文： ** 手術的；外科手術治療的****

84. ****operative treatment vs non-operative treatment****

operative

手術治療

non-operative

非手術 / 保守治療

85. **preoperative / pre-operative = 手術前的**

Examples:

- * preoperative assessment
- * preoperative fasting
- * preoperative imaging

86. **postoperative / post-operative = 手術後的**

Examples:

- * postoperative pain
- * postoperative care
- * postoperative infection

87. **perioperative = 圍手術期的**

Covers the period around surgery, typically including aspects of:

- * preoperative
- * intraoperative
- * postoperative

care.

88. **intraoperative = 手術中的**

Used for events/monitoring occurring during surgery.

R. OPERABLE / INOPERABLE — MEDICINE

89. **operable = suitable/possible to treat surgically (可手術的) **

Practice 20

> “The tumour was considered operable.”

「該腫瘤被認為是**可以進行手術的**。」

This does not necessarily mean surgery is definitely the best option—only that surgical treatment/removal is considered feasible in the relevant context.

90. **inoperable = not suitable/possible to treat surgically (無法手術的) **

> “The tumour was initially considered inoperable.”

「該腫瘤最初被認為**無法進行手術治療**。」

Reasons can differ:

- * location
- * extent
- * patient condition
- * technical feasibility

depending on case.

S. MILITARY

91. **operate = carry out military activity (軍事行動) **

> “The unit operated behind enemy lines.”

「該部隊在敵後**執行任務 / 活動**。」

92. **military operation = 軍事行動**

Can refer to:

- * combat
- * logistics
- * reconnaissance
- * humanitarian military activity

depending on context.

93. **combat operation = 作戰行動**

94. **special operation = 特別行動**

A broad military/security label; exact meaning depends on organization/context.

95. **theatre of operations = 作戰區域**

Military term for geographic area where operations are conducted.

96. **operational area = 行動區 / 作戰區**

T. POLICE / INTELLIGENCE

97. **police operation = 警方行動**

> “Police carried out an operation overnight.”

「警方通宵進行了一次**行動**。」

98. **undercover operation = 臥底 / 秘密行動**

99. **surveillance operation = 監視行動**

100. **sting operation = 誘捕 / 設局執法行動**

Especially AmE law-enforcement terminology.

Exact legal practices vary.

101. **operative — noun = agent/person carrying out covert or organized work (特工 / 行動人員) **

Practice 21

> “The intelligence operative worked overseas.”

「該名情報**特工 / 行動人員**在海外工作。」

Common:

* intelligence operative

* political operative

* covert operative

102. **political operative**

In American political language:

> person working strategically for a political organization/campaign.

This is descriptive of a role, not necessarily covert.

103. **operative — worker (較舊 / 英式) **

Historically/industrially:

> factory operative

= 工廠工人 / 操作員

Still appears in some occupational titles.

U. MATHEMATICS

104. **operation = mathematical process (運算) **

Examples:

- * addition
- * subtraction
- * multiplication
- * division

Practice 22

> “Multiplication is a mathematical operation.”

「乘法是一種數學**運算**。」

105. **arithmetic operation = 算術運算**

106. **binary operation = 二元運算**

In mathematics, an operation taking two inputs/operands under the specified definition.

107. **unary operation = 一元運算**

Takes one operand/input.

108. **order of operations = 運算次序**

Rules specifying the order in which mathematical operations are performed.

109. **inverse operation = 逆運算**

Example:

- * addition $\leftrightarrow$ subtraction
- * multiplication $\leftrightarrow$ division

under ordinary arithmetic interpretation.

V. OPERATOR — MATHEMATICS

110. **operator = symbol/function performing an operation (運算子) **

Examples:

* `+`

* `-`

* differentiation operator

* linear operator

**繁體中文： ** 運算子 / 算符 **

111. **linear operator = 線性算子**

Major linear-algebra/functional-analysis term.

112. **differential operator = 微分算子**

113. **operator acts on an operand/function**

Formal mathematical language:

> “The operator acts on the function.”

「該算子**作用於**這個函數。」

This is another technical echo of **operate on**.

W. COMPUTING — OPERATION

114. ****operation = computer/system action (操作 / 運算) ****

Examples:

- * read operation
- * write operation
- * delete operation
- * network operation
- * database operation

115. ****input/output operation = 輸入 / 輸出操作****

Often abbreviated:

> I/O operation.

116. ****atomic operation = 原子操作****

Computing/concurrency:

> operation treated as indivisible for the relevant concurrency model.

117. ****blocking / non-blocking operation****

Software/network terminology describing whether execution waits for completion.

118. ****asynchronous operation = 非同步操作****

X. OPERATING SYSTEM

119. ****operating system = 作業系統 / 操作系統****

A major computing compound.

An operating system manages computer resources and provides services/interfaces for software.

Examples of functions include:

- * process management
- * memory management
- * file systems
- * device management
- * security/access

120. **OS = operating system**

Common abbreviation.

121. **operating-system level**

= 作業系統層面.

Y. PROGRAMMING — OPERATOR

122. **operator = symbol/construct specifying an operation (運算子)**

Examples:

- * arithmetic operators
- * comparison operators
- * logical operators
- * assignment operators

123. **arithmetic operator**

Examples:

`+ - \* /`

124. **comparison/relational operator**

Examples:

`==`, `<`, `>`

depending on language.

125. ****logical operator****

Examples:

* AND

* OR

* NOT

or language-specific equivalents.

126. ****bitwise operator****

Operates on individual bits.

127. ****operator precedence = 運算子優先次序****

Determines which operators bind/evaluate first under language rules.

128. ****operator overloading = 運算子多載 / 重載****

Programming feature allowing operators to have type-specific behavior.

Z. SEARCH / DATABASE OPERATORS

129. ****search operator = 搜尋運算子 / 搜尋指令****

Examples can include operators restricting:

* site

- * file type
- * logical combinations

depending on search system.

130. **Boolean operator = 布林運算子**

AND / OR / NOT.

Used in:

- * search
- * logic
- * databases
- * programming

AA. OPERATOR = PERSON WHO CONTROLS EQUIPMENT

131. **operator = person who operates machinery/equipment (操作員) **

Practice 23

> “The crane operator must be properly trained.”

「起重機**操作員**必須受過適當訓練。」

Common:

- * machine operator
- * crane operator
- * forklift operator
- * camera operator

132. **camera operator = 攝影機操作員**

Person operating camera equipment in film/TV production.

133. **computer operator = 電腦操作員**

Historically/currently in certain IT operations roles.

134. **equipment operator = 設備操作員**

AB. OPERATOR = COMPANY PROVIDING A SERVICE

135. **operator = company/entity running a service or facility (營運商) **

Practice 24

> “The rail operator announced a timetable change.”

「鐵路**營運商**宣布時間表有變。」

Common:

- * network operator
- * transport operator
- * hotel operator
- * airport operator
- * facility operator

136. **mobile network operator = 流動網絡營運商**

137. **telecom operator = 電訊營運商**

138. **tour operator = 旅行團營辦商 / 旅遊營運商**

Company that organizes/packages travel services.

Not the same as:

> travel agent

although businesses may perform both roles.

139. **owner-operator = 自營業主 / 自營司機等**

Person who both:

- * owns;

- * personally operates

a business/vehicle.

Especially common in trucking and small businesses.

AC. TELEPHONE OPERATOR

140. **operator = person who connects/assists telephone calls (接線員) **

Historically very common:

- > telephone operator

- > switchboard operator

「電話**接線員**」

Still used in certain institutional/customer-service contexts.

141. **Operator, please... **

Traditional telephone expression.

Historical/cultural importance.

AD. **SMOOTH OPERATOR**

142. **smooth operator = socially skilful, confident person, sometimes manipulatively so (八面玲瓏的人) **

Informal idiom.

> “He's a smooth operator.”

Depending on tone:

「他是個很**圓滑 / 很會處事的人**。」

May suggest:

* charm

* confidence

* cunning

* manipulation

AE. **OPERATOR ERROR**

143. **operator error = mistake by the person using equipment (操作失誤) **

> “The failure was caused by operator error.”

「故障由**操作人員失誤**造成。」

Can be neutral/technical, though sometimes used dismissively.

AF. **OPERATIONAL**

144. **operational = functioning and ready for use (可運作的) **

Practice 25

> “The new system is now fully operational.”

「新系統現在已經完全**投入運作 / 可正常使用**。」

This is a very important adjective.

145. **fully operational = 全面投入運作**

146. **become operational = 開始投入運作**

> “The new terminal becomes operational next year.”

「新航站樓明年**投入運作**。」

147. **operational readiness = 運作 / 戰備準備程度**

Can apply to:

- * military units
- * emergency systems
- * technical infrastructure

148. **operational capability = 實際運作能力**

AG. OPERATIONAL = RELATING TO DAY-TO-DAY OPERATIONS

149. **operational = relating to practical running of an organization (營運上的) **

Examples:

- * operational costs
- * operational issues
- * operational efficiency
- * operational staff

150. **operational efficiency = 營運效率**

151. **operational problem = 營運問題**

A problem affecting day-to-day functioning.

152. **operational risk = 營運風險**

In finance/business risk management, commonly refers to risk of loss/disruption arising from failures of:

- * processes
- * people
- * systems
- * external events

Exact regulatory definitions vary.

AH. OPERATIONAL VS STRATEGIC / TACTICAL

153. **operational = concerned with implementation/day-to-day execution**

Common management contrast:

- * strategic = long-term direction
- * tactical = intermediate methods
- * operational = practical execution

The boundaries vary by organization.

AI. **OPERATIONALLY**

154. **operationally = from the viewpoint of practical operation (在營運 / 實際操作上) **

Practice 26

> “The plan is technically possible but operationally difficult.”

「這個方案技術上可行，但在**實際營運上**很困難。」

155. **operationally independent**

= 在營運 / 行動上獨立.

AJ. **OPERATIVE**

This family member has several independent meanings.

156. **operative = functioning / having effect (有效的 ; 起作用的) **

Formal:

> “The agreement remains operative.”

「協議仍然**有效 / 生效中**。」

157. **operative date = 生效日期**

The date from which a rule/agreement/provision has effect.

158. **operative provision/clause = 生效 / 具實際效力的條款**

Legal/formal.

159. **operative part = 實際有約束 / 效力的部分**

AK. **OPERATIVE WORD**

160. **the operative word = the key/significant word (關鍵字眼) **

A very important idiom.

Practice 27

> “The operative word is ‘temporary’.”

「這裏的**關鍵字眼**是『臨時』。」

Meaning:

> this particular word controls or reveals the important meaning.

161. **operative phrase = 關鍵語句**

Less common but analogous.

AL. OPERATION OF LAW

162. **by operation of law = automatically because of legal rules (依法自動產生) **

A major legal expression.

Broadly:

> a legal consequence occurs because the law itself produces it, rather than because someone specifically chooses/contracts for it.

繁體中文： **因法律的作用 / 依法自動**

163. **operation of a statute/rule = 法律 / 規則的作用或適用**

> “The right arises through operation of law.”

「該權利因**法律作用而產生**。」

AM. OPERATE = HAVE AN EFFECT / ACT UPON

164. **operate on/upon something = exert an effect (作用於) **

Formal/scientific/older usage:

> “Several forces operate on the structure.”

「數種力量**作用於**該結構。」

Modern English often prefers:

> act on.

165. **factors operate to produce a result = factors work to cause it**

Formal:

> “Several factors operate to reduce demand.”

「數個因素共同**發揮作用，使需求下降**。」

166. **law/rule operates to X = has the effect of X**

Legal/formal:

> “The rule operates to prevent double recovery.”

「該規則的作用是**防止重複獲得賠償**。」

167. **operate in favour of / against someone**

Formal:

> factors operate in someone's favour

= 各項因素對某人**有利**。

AN. RESEARCH — OPERATIONAL DEFINITION

168. **operational definition = definition specifying how something is measured/identified (操作性定義)**

A major research-methods term.

Example:

> defining “stress” by a specified questionnaire score.

The concept is translated:

> **操作性定義 / 操作型定義**

depending on local academic convention.

169. **operationalise / operationalize a concept = turn abstract concept into measurable variables (操作化) **

Practice 28

> “Researchers must operationalize the concept of social trust.”

「研究人員必須把『社會信任』這個概念**操作化**。」

Meaning:

> specify observable/measurable indicators.

170. **operationalisation / operationalization = 操作化**

The process of turning theoretical concepts into measurable variables/procedures.

AO. OPERATIONS RESEARCH

171. **operations research = 運籌學**

An interdisciplinary mathematical/analytical field using methods such as:

- * optimization
- * modelling
- * probability
- * simulation

to improve decision-making.

172. **operational research = 運籌學**

Common British term for:

> operations research.

Often abbreviated **OR**.

AP. ELECTRONICS — OPERATIONAL AMPLIFIER

173. **operational amplifier = 運算放大器**

Common abbreviation:

> **op-amp**

An electronic amplifier building block extensively used in analog circuits.

The word **operational** historically relates to its use in analog mathematical operations.

AQ. CHEMICAL / INDUSTRIAL ENGINEERING

174. **unit operation = 單元操作**

A fundamental physical processing step in chemical/process engineering.

Examples include:

- * distillation
- * filtration
- * drying
- * heat transfer

depending on classification.

175. **operating unit = operating component/facility**

Meaning varies by industry.

176. **operating procedure = 操作程序**

177. **standard operating procedure = 標準作業程序**

Common abbreviation:

> ****SOP****

A documented standardized way of performing a recurring task/process.

178. ****operating instructions = 操作說明****

179. ****operating manual = 操作手冊****

AR. ENGINEERING — OPERATING POINT

180. ****operating point = system's actual steady-state condition (工作點) ****

Used in:

- * electronics
- * engines
- * pumps
- * control systems

Exact variables depend on system.

181. ****normal operating point****

= 正常工作點.

182. ****operating temperature = 工作 / 操作溫度****

183. ****operating pressure = 工作壓力****

184. ****operating speed = 運轉速度****

AS. OPERABILITY

185. **operability = ability/ease with which a system can be operated effectively (可操作性) **

Used in:

- * engineering
- * process design
- * software
- * industrial systems

****繁體中文： ** 可操作性；操作性能****

186. **operability study**

An engineering study assessing whether systems can be operated safely/effectively under expected conditions.

187. **HAZOP — hazard and operability study**

A structured process-safety analysis method:

> Hazard and Operability Study.

****繁體中文： ** 危害與可操作性分析 / 研究****

Specialist engineering term.

AT. OPERABLE — GENERAL/TECHNICAL

188. **operable = capable of being operated/worked (可操作的) **

> “The controls must remain operable.”

「控制裝置必須保持**可操作狀態**。」

Less common in everyday English than:

> working / functional.

189. **inoperable = unable to function/be operated (無法操作 / 失靈) **

Outside medicine:

> “The mechanism became inoperable.”

「機械裝置變得**無法操作**。」

AU. INTEROPERABILITY

190. **interoperate = systems work together (互操作) **

> “The two systems must interoperate.”

「兩套系統必須能夠**互操作**。」

191. **interoperable = able to work with other systems (可互操作的) **

Common in:

- * software
- * communications
- * healthcare systems
- * defence systems

192. **interoperability = ability of systems to exchange/use information or work together (互操作性) **

A major technology/standards term.

This is an established extended family rather than a bare **operate** sense.

AV. IMPORTANT FIXED CONSTRUCTIONS

193. ****operate independently****

> function/run without dependence on another entity.

「獨立運作」

194. ****operate jointly****

> run something together.

「聯合營運」

195. ****operate successfully****

> function/conduct activity successfully.

196. ****operate safely****

Machine/company/process may operate safely.

197. ****operate efficiently****

= 有效率地運作 / 營運.

198. ****operate continuously****

= 連續運轉.

199. ****operate on a first-come, first-served basis****

Here:

> operate on a ... basis

= 按某原則 / 模式運作。

200. **operate according to a principle**

> “The system operates according to a simple rule.”

「系統按照一項簡單規則**運作**。」

AW. IMPORTANT DISTINCTIONS

operate vs work — machine

work

everyday word meaning function.

operate

more formal/technical.

> “The printer isn't working.”

more conversational than:

> “The printer is not operating.”

operate vs run — machine/system

run

very common for engines/software/systems.

operate

more formal, emphasizing functioning.

> run the engine

> operate the machinery

Large overlap.

operate vs control

operate

use controls to make equipment function.

control

determine/direct its behavior.

A person may operate a machine without controlling every automatic process inside it.

operate vs use

use

very broad.

operate

specifically implies controlling machinery/system functionality.

> use a calculator

> operate a crane

operate vs manage

manage

organize/direct people/resources.

operate

run the actual business/service/system.

A manager may manage a company while operations staff operate facilities.

operate vs run — business

Very close:

> run a restaurant

> operate a restaurant

Run is more everyday.

Operate is more formal/businesslike.

operate vs own

Critical:

own

possess legally/economically.

operate

run/manage.

The owner and operator can be different entities.

Hence:

> owner-operator

specifically combines both roles.

operator vs owner

operator

entity actually running the service/facility.

owner

entity possessing it.

A hotel may be:

> owned by Company A
> operated by Company B.

operate vs administer

administer

manage/implement a program, organization, rule, or treatment.

operate

run the activity/system itself.

operate vs conduct

conduct

carry out an activity.

operate

can mean run a continuing organization/system.

> conduct an investigation
> operate a business

operate vs function

function

serve/work in an intended way.

operate

can mean function **or** cause/control functioning.

> "The device functions normally."
> "The device operates normally."

Nearly equivalent.

But:

> “She operates the device.”

cannot normally be replaced by:

> *She functions the device.* ❌

operation vs function

operation

actual process/activity of working.

function

intended role/purpose OR act of working.

> function of the heart

> operation of the heart

different emphasis:

* function = what it does

* operation = how/that it works.

operation vs process

process

series of steps/changes.

operation

specific action/activity or functioning.

A process may contain several operations.

operation vs procedure

procedure

prescribed sequence/method.

operation

activity itself.

In medicine, however, **operation** and **surgical procedure** overlap strongly.

operation vs activity

activity

very broad.

operation

often organized, goal-directed, and coordinated.

> rescue operation

sounds more structured than:

> rescue activity.

operating cost vs operating expense

They overlap heavily, but accounting presentations can distinguish them.

operating cost

broad cost of carrying out operations.

operating expense

accounting expense from normal operations, often excluding particular production costs depending on presentation.

Do not mechanically treat them as identical in every financial statement.

operating cost vs overhead

As above:

- > overhead = generally indirect cost
- > operating cost = broader running cost.

operational vs operating

operating

often directly connected with the act/system of operation:

- * operating cost
- * operating system
- * operating temperature

operational

often means:

- * functioning/ready; or
- * relating to practical operations.

- > operational readiness
- > operational problem

Collocations are not freely interchangeable.

operational vs operative

operational

functioning/ready or related to operations.

operative

effective/in force, surgical, or key/significant.

Compare:

> operational system
> operative clause
> operative treatment

operational vs strategic

strategic

long-term goals/direction.

operational

practical implementation/day-to-day execution.

operative vs effective

operative

formally in force/working.

effective

actually produces the intended result.

A rule can be operative yet not especially effective.

operative vs operating

operating room

手術室

not normally:

> *operative room* ✗

operative treatment

手術治療

not normally:

> *operating treatment* ❌

Collocation matters.

operable vs operational

operable

capable of being operated; in medicine capable of surgery.

operational

already functioning/ready for use.

A machine may be:

> operable but not currently operational.

inoperable vs non-operational

inoperable

cannot be operated/functions no longer possible; or cannot be treated surgically.

non-operational

not currently operating/ready.

Something can be temporarily non-operational without being inoperable.

operate on vs operate

Medical:

> operate **on** a patient.

Machine:

> operate **a** machine.

This is an important grammar contrast.

AX. GRAMMAR

201. **operate — intransitive**

> “The system operates automatically.”

= 系統自動運作。

202. **operate + object**

> operate a machine

> operate a business

> operate a service

Meaning depends strongly on object.

203. **operate on + patient/body part**

> operate on a patient

> operate on the heart

= 動手術。

204. **operate in + place/industry**

> operate in Hong Kong

> operate in retail

= 在某地 / 行業營運。

205. **operate from + base/location**

> operate from an office

= 以該地點為基地營運。

206. **operate under + name/licence/rules**

> operate under a licence

= 持牌營運。

207. **operate as + role**

> operate as a charity

= 以慈善機構形式運作。

208. **operate at + level/condition**

> operate at full capacity

> operate at a loss

> operate at high pressure

209. **operate between + locations**

Especially transport:

> trains operate between A and B.

210. **operate to + verb — formal**

> “The rule operates to prevent abuse.”

= 該規則的作用是防止濫用。

Formal/legal.

AY. COUNTABILITY

operation — functioning/activity

Can be uncountable:

> operation of the machine

or countable:

> a rescue operation

> three operations

operation — surgery

Countable:

> an operation

> two operations

operations — business

Usually plural:

> business operations

> daily operations

operator

Countable:

> an operator

> several operators

operability

Usually uncountable:

> improve operability.

AZ. FAMILY MAP

Family member	Main modern meanings	
-----	-----	
operate — function	work/run	
operate — control	control/use machinery	
operate — business	run/manage a business or organization	
operate — service	provide/run a service, route, facility	
operate — location	conduct activities in/from a place or market	
operate — surgery	perform surgery on someone	
operate — effect	formally act/have an effect	
operation	functioning; organized activity; surgery; mathematical/computing action	
operations	day-to-day business activities; military/police activities	
operating	relating to ongoing operation: costs, systems, conditions, surgery	
operator	equipment user; service-running company; telephone operator; mathematical/computing operator	
operational	functioning/ready; relating to practical operations	
operationally	in practical/operational terms	
operative — adjective	effective/in force; surgical	
operative — noun	agent/worker	
operable	capable of operation or surgery	
inoperable	unable to be operated / surgically treated	
operability	ease/ability of operating successfully	
operationalize	turn an abstract concept into measurable procedures	
interoperable	capable of functioning together with another system	

Master comparison

Form / sense	MCQ-ready Traditional Chinese meaning
-----	-----

operate — machine functions	機器 / 系統本身正常運作	
operate a machine	人控制、使用機械；操作	
operate a business	經營 / 營運一門生意	
operate a restaurant/hotel	負責該商業場所的日常經營	
operate a service	營辦、提供某項持續服務	
service operates	服務按時間 / 路線正常提供	
operate in a market/country	在某市場 / 地方開展業務活動	
operate from X	以 X 作為營運基地	
operate under a name/licence	以某名稱 / 牌照制度營運	
operate as X	以某種身分 / 機構形式運作	
operate at a profit/loss	盈利 / 虧損經營	
operate at full capacity	以最大容量運作	
operating cost	維持日常業務運轉所產生的營運成本	
fixed/variable operating cost	固定 / 變動營運成本	
operating expense / OPEX	日常業務活動產生的營運開支	
operating income/profit	主要業務活動產生的營業收益 / 利潤	
operating margin	營業利潤相對收入的比率	
operating cash flow	日常營運活動產生的現金流	
operating leverage	固定營運成本令利潤對收入變化產生放大效應的營運槓桿	
business operations	公司日常營運活動	
operations manager	負責日常流程 / 服務營運的經理	
operation of a machine	機器的運作 / 運作方式	
in operation	正在使用 / 運作中	
put into operation	正式投入運作	
cease operations	停止業務 / 行動	
rescue/search operation	有組織的救援 / 搜查行動	
operation — surgery	手術	
operate on someone	為某人進行外科手術	
operating room/theatre	手術室	
operative treatment	手術治療	
preoperative	手術前的	
postoperative	手術後的	
perioperative	圍繞手術前、中、後期的	
operable tumour	技術 / 臨床上可以考慮手術治療的腫瘤	
inoperable tumour	無法 / 不適合手術治療的腫瘤	
military operation	有組織的軍事行動	
police operation	警方有計劃的執法行動	
operative — agent	情報 / 政治等行動人員、特工	
mathematical operation	加、減、乘、除等數學運算	
operator — mathematics	對數值 / 函數等執行運算的算子 / 運算子	
computer operation	電腦系統執行的一項操作	
operating system	管理電腦硬件資源及提供軟件運行環境的作業系統	
programming operator	程式語言中的算術、比較、邏輯等運算子	
operator — machine worker	操作機器 / 設備的人員	
operator — service company	實際營運交通、網絡、設施等服務的營運商	

tour operator	組織 / 包裝旅遊產品的旅遊營辦商	
telephone operator	電話接線員	
smooth operator	善於應對、圓滑甚至帶操縱意味的人	
operator error	因操作人員失誤造成的錯誤	
operational — ready	已準備好並能正常投入使用的	
fully operational	已全面投入正常運作	
operational — business	與日常營運實務有關的	
operational risk	因人員、流程、系統或外部事件等引起的營運風險	
operationally	從實際營運 / 執行角度而言	
operative — in force	正在生效、具實際效力的	
operative date	法規 / 協議等開始具效力的日期	
operative word	決定句意或論點的關鍵字眼	
by operation of law	法律效果自動因法律規定產生	
operate on/upon — formal effect	對某事物發生作用 / 影響	
operational definition	用具體觀察 / 測量方式界定抽象概念的操作性定義	
operationalize	把抽象概念轉化成可以觀察 / 測量的變數	
operations research	用數學模型、最佳化等方法改善決策的運籌學	
operational amplifier	電子學中的運算放大器	
unit operation	化工 / 流程工程中的單元操作	
standard operating procedure	標準作業程序 (SOP)	
operating point	工程系統實際穩態工作的工作點	
operability	系統是否容易、安全、有效操作的可操作性	
interoperability	不同系統能交換資訊並協同工作的互操作性	

For your original passage

The relevant phrase is:

> “...the private tutoring industry ****externalise its operating costs**** while avoiding responsibility.”

The exact meaning is:

****operating costs** = 如果補習業務正常在正式場所營運，本來需要自行承擔的日常經營成本

For example, a tutor or centre might ordinarily have to bear:

- * rent for teaching premises;
- * utilities;
- * cleaning;
- * equipment;
- * administration;
- * other venue-related running costs.

By occupying restaurant tables instead, the writer argues that some of those costs are effectively shifted onto the restaurant.

So:

> **externalise its operating costs**

≈

> **把原本屬於自己營運所需承擔的成本轉嫁給其他企業或社會**

There is also a useful connection to your earlier word **overhead**:

> **operating costs** = 經營業務的整體日常成本

> **overhead** = 通常指不能直接歸到某一件產品 / 服務上的間接成本

Therefore **rent** can be both an operating cost and, depending on the accounting context, **overhead**, but the two expressions are not identical.

For the core database, I would keep **7 major base-verb branches** for **operate**:

1. **function/work** — a machine or system operates
2. **control/use machinery** — operate a machine
3. **run/manage a business or organization**
4. **provide/run a service, route, or facility**
5. **conduct activities in/from a place, industry, or market**
6. **perform surgery** — operate on someone
7. **formal: exert an effect / act upon something**

Then the whole family should separately index **operation, operations, operating, operator, operational, operative, operable/inoperable, operability, operationalize**, and **interoperability**, because those forms have developed major independent branches across **business/accounting, medicine, military/police work, mathematics, computing, engineering, law, research methods, and technology**.